Kathan Desai

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SUMMARY: Skilled Engineer with 3 years of professional experience seeking full-time opportunity in the field of RF & Microwave Circuit Design.

EDUCATION

Master of Science in Electrical Engineering, University of South Florida, Tampa, FL. **Expected May 19**
GPA: 3.66/4.00
Courses: RF/Microwave Circuit-1, Wireless Circuit/System Lab, Wireless Network Architecture, Digital Communication System, Wireless Sensor Network, Digital Signal Processing-1&2.

Bachelor of Engineering in Instrumentation and Control, Gujarat Technological University, India. **May 14**
GPA: 7.65/10

SKILLS

- Environment: Agilent ADS, MATLAB, Proteus, PSpice.
- Programming: C, C++.
- Application: Microsoft Office Suite.
- Tools: Digital Oscilloscope, Spectrum Analyzer, Vector Network Analyzer, Signal Generator.

KEY PROJECTS:

Impedance Matching Circuit Design, USF. **Oct 2018**
- Designed a distributed microstrip impedance matching network to match a non-50 Ohm load to a 50 Ohm using an open-circuited stub with a maximum of 10 dB return loss.
- Calculated the unknown load using Smith Chart Utility tool. Used an Open Circuited Stub for matching network for its ease of fabrication than the short-circuited stub.
- Line Cal function was used to calculate the length of the transmission line and the stub and with the help of proper tuning maximum of 10 dB return loss was obtained. This was an individual project where the bandwidth of 12.1% was obtained within the size constraint.

Coupled line Bandpass Filter, USF. **Nov 2018**
- Designed third order Bandpass Filter with a center frequency of 2.45 GHz using Microstrip coupled lines on FR4 substrate with a good Figure of Merit (FOM). A 0.5 dB Chebyshev Filter with 10% bandwidth was taken into consideration with the approach of Insertion Loss method.
- Linecalc was used to find the length, width, and separation of the coupler. Tuning of length, width, and separation was carried out to obtain FOM of 12.795. My design was selected for milling purpose for the highest FOM within the group. The Circuit was tested on VNA to compare results.

Echo Cancellation Using LMS algorithm, USF. **Feb 18 – May 18**
- The goal of this project was to develop an adaptive LMS algorithm for noise cancellation in MATLAB software.
- Defined the input signal, the desired signal, and the output signal to a system. Developed a code with updating weights of the filter and suitable step size for better convergence rate. An individual project where Adaptive nature was studied.

Security Issues and Vulnerability in Connected Car Systems, USF. **Oct 18 – Dec 18**
- Developed a MATLAB code for encryption and decryption for a valid communication between ECUs using cryptography in today's connected vehicle technology.

EXPERIENCE

1.Enviro Control Associates (I) Pvt Ltd, Surat, India.
Plant Supervisor **Dec 16 – Apr 17**
- Guided and managed workforce of over 100 individuals including specialists, engineers, and technicians.
- Executed the Project related work for the plant augmentation. Analyzed contractual requirements, determined the project phases and objectives by conferring with management.
- Coordinated with contractors and management to resolve issues for uninterrupted work. Maintained coherency and enlightened government officials with the progress of project work.

Instrumentation Engineer **Aug 14 – Nov 16**
- Successfully resolved the frequent breakdown of the soft starter by suggesting reducing the switching of SS in one cycle.
- Applied PM/PdM activities to improve mean time between failures.